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E-203 POWER DISTRIBUTION CENTER (PDC) TRANSFORMER

1. SCOPE OF WORK

The work shall include the design, supply, factory testing, delivery to the site, installation, commissioning and testing of all power distribution center transformers in accordance with Part 4, Subsection 1.2 Summary of Work.

2. CODES AND STANDARDS

The Contractor shall comply with the requirements of Part 4, Subsection 2.5 Codes and Standards.

The following codes and standards are specifically applicable to the design, manufacture and testing of the equipment included in this specification.

2.1 National

- PUIL : Peraturan Umum Instalasi Listrik - Indonesia (General Regulation of Electrical Installation - Indonesia)
- SPLN : Standard Perusahaan Umum Listrik Negara

2.2 International Electrotechnical Commission

- IEC 60076 : Power Transformers
- IEC 60296 : Specification for unused mineral oil for transformers and switchgear
- IEC 60354 : Loading guide for oil immersed transformers
- IEC 60439 : Low-voltage switchgear and controlgear assemblies
- IEC 60529 : Degrees of protection provided by enclosures (IP Code)
- IEC 60726 : Dry-type power transformers

2.3 National Electrical Manufacturers Association

- NEMA TR1 : Transformers, Regulators, and Reactors

3. OPERATING AND DESIGN CONDITIONS

3.1 Environment

All equipment shall be suitable for operation in a combined cycle power plant on the seacoast. Refer to Part 4, Subsections 1.1, 2.14 and 2.16 for the meteorological, seismic, and environmental conditions on the site.

3.2 Equipment Ratings

- (1) Voltage ratio at nominal taps : 6.3 kV / 400 V
- (2) Frequency and number of phases : 50 Hz, 3 phase.
- (3) Off load taps : $\pm 2 \times 2.5 \%$
- (4) Capacity : *) kVA AN/AF or *) kVA ONAN/ONAF depending on transformer type.
- (5) Impedance : **)
- (6) Maximum rated voltage applied of the high voltage winding : 6.6 kV.
- (7) Connections : Dyn1 configuration, 6.3 kV winding delta connected, 400 V winding wye connected solidly grounded
- (8) Insulation Level:

	6.3 kV winding	400 V winding
a. Impulse Voltage	: 60 kV	----
b. Short duration power frequency	: 20 kV	3 kV
- (9) Insulation Class:
 - a. Indoor dry type : Class F or better
 - b. Outdoor liquid filled type : Class A or better

Note: *) The Contractor shall determine size of their PDC Transformers to suit not only static and dynamic load requirement but also suit for essential load of other unit and giving 15 % spare capacity. Sizing calculations shall be submitted for approval by the Owner.

3.3 PDC Transformer Designations

All PDC Transformers shall preferably be indoor, dry type, except for hazardous areas locations.

4. DETAILED REQUIREMENTS

4.1 General Requirements

Transformers shall be manufactured or assembled in Indonesia. All indoor transformers shall be of the ventilated dry type. All outdoor transformers shall be of the weatherproof liquid filled type.

4.2 Structure and Housing

4.2.1 Indoor Transformers

Each indoor transformer shall be completely enclosed and provided with suitable louvers to protect it from falling dirt and dripping liquids. The transformer shall have a minimum degree of protection of IP 31.

Rigid angle iron or channel iron supporting frames or structures shall be provided to support the transformer, buses, connections and accessories in the housing.

4.2.2 Outdoor Transformers

All outdoor liquid filled transformers shall have a degree of protection of IPW 55.

Oil filling level indications shall be engraved on the inside and outside of all oil-filled chambers. All transformers shall be equipped with oil filling and drain valves. A sealed tank type of oil preservation system is preferred.

Equipment enclosures, supporting frames, bases and all non-current carrying parts shall be bonded together and grounded by means of the ground bus.

4.3 Bushings, Insulators, Supports and Fittings

Bushings, insulators and supports shall be provided for mounting and bracing the conductors in the transformer housing and shall be braced so as to prevent the displacement of any part due to stresses incident to short circuits.

Each power distribution center shall be equipped with a copper ground bus. The minimum ground bus size shall be 50 x 7 mm. The ground bus shall be

arranged so that it will form an electrically continuous bus with the power distribution center switchgear ground bus when the transformer and switchgear are mounted adjacent to each other.

4.4 Terminal Connections

The low voltage terminals shall be designed for close-coupled connection to PDC switchgear. The neutral of the low voltage shall be connected to a bushing. Outdoor transformers shall be designed for connection to non-segregated bus duct or cable as required.

For indoor transformers, a provision shall be made for detaching the connections from the cable and switchgear and moving the transformer in a direction perpendicular to the lineup.

4.5 Off-Circuit Tap Changer

The transformer shall be equipped with a terminal board or convenient terminal for changing tap connections when the transformer is not excited. It shall not be necessary to drain the oil of outdoor transformers to change the taps.

4.6 Windings

Indoor transformer windings shall have Class F or better insulation. The maximum winding temperature rise shall be 80°C at the transformer full nameplate rating.

Liquid-filled transformer windings shall have a maximum temperature rise of 65°C at the transformer full nameplate rating.

The transformers and their components shall be capable of being temporarily overloaded in accordance with IEC 60354.

The transformer insulation shall be able to withstand the switching surges produced by the medium voltage switchgear. The winding assembly of all dry type transformers shall be non-hygroscopic.

4.7 Temperature Indicator and Relays

The transformers shall be provided with a dial type temperature indicator to show hot spot temperature and a temperature alarm relay will close its contacts on high temperature.

The transformers shall be equipped with a temperature relay to operate the cooling fans when the transformer hot spot temperature exceeds the full load rated temperature.

4.8 Forced Cooling

Each transformer shall be equipped with cooling fans. The fans shall be automatically started and stopped by the temperature switch of Clause 4.7. It shall be possible to operate the transformer at 1 1/3 times its naturally cooled rating without exceeding the transformer's rated temperature rise.

4.9 Core

The transformer core shall be made of high quality grain oriented steel. The core shall be laminated and each lamination insulated by a suitable varnish to avoid eddy current losses. The core shall be supported on insulators and grounded with a removable core grounding link.

4.10 Insulating Oil

Insulating oil or fluid shall be as required by Technical Specification E-001.

4.11 Noise Level

The transformer shall be designed and constructed so that the noise level will not exceed the value listed in NEMA TR1.

4.12 Small Wiring

Control wiring shall be 2.5 mm² except where larger size conductors are needed for current carrying requirements. Current transformer leads shall be 6 mm² or larger. The conductors shall be stranded copper.

4.13 Nameplates

The transformer shall be provided with a nameplate in accordance with IEC 60726 or 60076, as applicable.

4.14 Current Transformer

The 400 Volt neutral connection shall be equipped with a suitable rated current transformer meeting the requirements of Technical Specification E-001. The current transformer shall be connected to shorting type terminal blocks and connected to the relays in the medium voltage breaker feeding the transformer.

5. PERFORMANCE, TESTING AND GUARANTEES

Type testing shall be as required by IEC 60726 for indoor transformers and IEC 60076 for outdoor transformers. If the type tests performed on similar transformers already done, then the certificate shall be submitted.

All routine tests required by IEC 60726 or 60076, as appropriate, shall be carried out. In addition, all transformers shall be impulse tested.

The Contractor shall provide test reports or calculations verifying the transformer's ability to withstand short circuits as specified herein.

6. QUALITY ASSURANCE/QUALITY CONTROL

Quality assurance/quality control shall conform to the requirements of Part 4, Section 2, Subsection 2.7. shall include, but not be limited to, the followings:

- (1) Design criteria and Engineering Calculations
- (2) Continuity and Wiring Tests
- (3) Insulation and High Voltage Tests
- (4) Packaging and Shipping
- (5) Welding

The documentation as specified shall include, but not limited to, the followings:

- (1) Material Reports
- (2) Inspection Reports
- (3) Shop Test Reports

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